

# The Riemann Integral

## Real Analysis

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Morales C. Ronny

Tagged Partitions

Riemann  
Integrability

Properties of  $\int_a^b$   
Improper Integrals

The Fundamental  
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## Tagged Partitions

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## Definition

If  $I := [a, b]$  is a closed, bounded interval in  $\mathbb{R}$ , then a **partition** of  $I$  is a finite, ordered set  $\mathcal{P} := (x_0, x_1, \dots, x_{n-1}, x_n)$  of points in  $I$  such that

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b.$$

The points of  $\mathcal{P}$  are used to divide  $I = [a, b]$  into non-overlapping subintervals

$$I_1 := [x_0, x_1], \quad I_2 := [x_1, x_2], \dots, \quad I_n := [x_{n-1}, x_n].$$

Often we will denote the partition  $\mathcal{P}$  by the notation  $\mathcal{P} = \{[x_{i-1}, x_i]\}_{i=1}^n$ . We define the **norm** (or **mesh**) of  $\mathcal{P}$  to be the number

$$\|\mathcal{P}\| := \max\{x_1 - x_0, x_2 - x_1, \dots, x_n - x_{n-1}\}.$$

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## Definition

If a point  $t_i$  has been selected from each subinterval  $I_i = [x_{i-1}, x_i]$ , for  $i = 1, 2, \dots, n$ , then the points are called **tags** of the subintervals  $I_i$ . A set of ordered pairs

$$\dot{\mathcal{P}} := \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$$

of subintervals and corresponding tags is called a **tagged partition** of  $I$ .

## Definition

If  $\dot{\mathcal{P}}$  is a tagged partition of  $I$ , we define the **Riemann sum** of a function  $f : [a, b] \rightarrow \mathbb{R}$  corresponding to  $\dot{\mathcal{P}}$  to be the number

$$S(f, \dot{\mathcal{P}}) := \sum_{i=1}^n f(t_i)(x_i - x_{i-1}).$$

## Definition

A function  $f : [a, b] \rightarrow \mathbb{R}$  is said to be **Riemann integrable** on  $[a, b]$  if there exists a number  $L \in \mathbb{R}$  such that for every  $\varepsilon > 0$  there exists  $\delta_\varepsilon > 0$  such that if  $\dot{P}$  is any tagged partition of  $[a, b]$  with  $\|\dot{P}\| < \delta_\varepsilon$ , then

$$\left| S(f, \dot{P}) - L \right| < \varepsilon.$$

The set of all Riemann integrable functions on  $[a, b]$  is denoted by  $\mathcal{R}[a, b]$ .

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*If  $f \in \mathcal{R}[a, b]$ , then the value of  $L$  is uniquely determined.*

## Definition

If  $f \in \mathcal{R}[a, b]$ , then the number  $L$  is called the **Riemann integral of  $f$**  over  $[a, b]$ . Instead of  $L$ , we will usually write

$$L = \int_a^b f \quad \text{or} \quad \int_a^b dx f(x).$$

*It should be understood that any letter other than  $x$  can be used in the latter expression, so long as it does not cause ambiguity.*

## Theorem

*If  $g$  is Riemann integrable on  $[a, b]$  and if  $f(x) = g(x)$  except for a finite number of points in  $[a, b]$ , then  $f$  is Riemann integrable and*

$$\int_a^b f = \int_a^b g.$$

## Theorem

Suppose that  $f$  and  $g$  are in  $\mathcal{R}[a, b]$ . Then:

a) If  $k \in \mathbb{R}$ , the function  $kf$  is in  $\mathcal{R}[a, b]$  and

$$\int_a^b kf = k \int_a^b f.$$

b) The function  $f + g$  is in  $\mathcal{R}[a, b]$  and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g.$$

c) If  $f(x) \leq g(x)$  for all  $x \in [a, b]$ , then

$$\int_a^b f \leq \int_a^b g.$$

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### Theorem (Boundedness Theorem)

*If  $f \in \mathcal{R}[a, b]$ , then  $f$  is bounded on  $[a, b]$ .*

### Theorem (Cauchy Criterion)

*A function  $f : [a, b] \rightarrow \mathbb{R}$  belongs to  $\mathcal{R}[a, b]$  if and only if for every  $\varepsilon > 0$  there exists  $\eta_\varepsilon$  such that if  $\dot{\mathcal{P}}$  and  $\dot{\mathcal{Q}}$  are any tagged partitions of  $[a, b]$  with  $\|\dot{\mathcal{P}}\| < \eta_\varepsilon$  and  $\|\dot{\mathcal{Q}}\| < \eta_\varepsilon$ , then*

$$\left| S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}}) \right| < \varepsilon.$$

## Theorem (Squeeze Theorem)

Let  $f : [a, b] \rightarrow \mathbb{R}$ . Then  $f \in \mathcal{R}[a, b]$  if and only if for every  $\varepsilon > 0$  there exist functions  $\alpha_\varepsilon, \omega_\varepsilon$  in  $\mathcal{R}[a, b]$  with

$$\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon \quad \text{for all } x \in [a, b],$$

and such that

$$\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon.$$

## Definition

A function  $\varphi : [a, b] \rightarrow \mathbb{R}$  is called a **step function** if  $[a, b]$  is the union of nonoverlapping intervals  $I_1, I_2, \dots, I_n$  such that  $\varphi$  is constant on each interval, that is,  $\varphi(x) = c_k$  for all  $x \in I_k$ ,  $k = 1, 2, \dots, n$ .

## Lemma

*If  $J$  is a subinterval of  $[a, b]$  having endpoints  $c < d$  and if  $\varphi_J(x) := 1$  for  $x \in J$  and  $\varphi_J(x) := 0$  elsewhere in  $[a, b]$ , then  $\varphi_J \in \mathcal{R}[a, b]$  and*

$$\int_a^b \varphi_J = d - c.$$

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### Theorem

*If  $\varphi : [a, b] \rightarrow \mathbb{R}$  is a step function, then  $\varphi \in \mathcal{R}[a, b]$ .*

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### Theorem

*If  $f : [a, b] \rightarrow \mathbb{R}$  is continuous on  $[a, b]$ , then  $f \in \mathcal{R}[a, b]$ .*

## Theorem (Additivity Theorem)

*Let  $f : [a, b] \rightarrow \mathbb{R}$  and let  $c \in (a, b)$ . Then  $f \in \mathcal{R}[a, b]$  if and only if its restrictions to  $[a, c]$  and  $[c, b]$  are both Riemann integrable. In this case*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

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### Corollary

*If  $f \in \mathcal{R}[a, b]$ , and if  $[c, d] \subseteq [a, b]$ , then the restriction of  $f$  to  $[c, d]$  is in  $\mathcal{R}[c, d]$ .*

## Corollary

*If  $f \in \mathcal{R}[a, b]$  and if  $a = c_0 < c_1 < \cdots < c_m = b$ , then the restrictions of  $f$  to each of the subintervals  $[c_{i-1}, c_i]$  are Riemann integrable and*

$$\int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f.$$

## Definition

*If  $f \in \mathcal{R}[a, b]$  and if  $\alpha, \beta \in [a, b]$  with  $\alpha < \beta$ , we define*

$$\int_{\beta}^{\alpha} f := - \int_{\alpha}^{\beta} f \quad \text{and} \quad \int_{\alpha}^{\alpha} f := 0.$$

## Theorem

If  $f \in \mathcal{R}[a, b]$  and if  $\alpha, \beta, \gamma$  are any numbers in  $[a, b]$ , then

$$\int_{\alpha}^{\beta} f = \int_{\alpha}^{\gamma} f + \int_{\gamma}^{\beta} f,$$

*in the sense that the existence of any two of these integrals implies the existence of the third integral and the validity of the previous equation.*

## Definition

If  $f$  is in  $\mathcal{R}[a, b]$  for every  $b > a$  and if the limit

$$\lim_{b \rightarrow \infty} \int_a^b f$$

exists in  $\mathbb{R}$ , then the **improper integral**

$$\int_a^\infty dt f(t)$$

is defined to be this limit.

## Theorem (Integral test)

Let  $f$  be a positive, decreasing function on  $\{t : t \geq 1\}$ . Then the series

$$\sum_{k=1}^{\infty} f(k)$$

converges if and only if the improper integral

$$\int_1^{\infty} dt f(t)$$

exists. In case of convergence, the partial sum  $s_n = \sum_{k=1}^n f(k)$  and the sum  $s = \lim(s_n)$  satisfy the estimate

$$\int_{n+1}^{\infty} dt f(t) \leq s - s_n \leq \int_n^{\infty} dt f(t).$$

## Theorem

*Let  $(f_n)$  be a sequence of functions in  $\mathcal{R}[a, b]$  and suppose that  $(f_n)$  converges uniformly on  $[a, b]$  to  $f$ . Then  $f \in \mathcal{R}[a, b]$  and*

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

## Theorem (Fundamental Theorem of Calculus (First Form))

*Suppose that there is a finite set  $E$  in  $[a, b]$  and functions  $f, F := [a, b] \rightarrow \mathbb{R}$  such that:*

- a)  $F$  is continuous on  $[a, b]$ ,*
- b)  $F'(x) = f(x)$  for all  $x \in [a, b] \setminus E$ ,*
- c)  $f$  belongs to  $\mathcal{R}[a, b]$ .*

*Then we have*

$$\int_a^b f = F(b) - F(a).$$

## Definition

If  $f \in \mathcal{R}[a, b]$ , then the function defined by

$$F(z) := \int_a^z f \quad \text{for } z \in [a, b],$$

is called the **indefinite integral** of  $f$  with **basepoint**  $a$ .

## Definition

Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . If there exists a constant  $K > 0$  such that

$$|f(x) - f(u)| \leq K|x - u|$$

for all  $x, u \in A$ , then  $f$  is said to be a **Lipschitz function** (or to satisfy a **Lipschitz condition**) on  $A$ .

## Definition

Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . We say that  $f$  is **uniformly continuous** on  $A$  if for each  $\varepsilon > 0$  there is a  $\delta > 0$  such that if  $x, u \in A$  are any numbers satisfying  $|x - u| < \delta$ , then  $|f(x) - f(u)| < \varepsilon$ .

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### Corollary

*If  $f : A \rightarrow \mathbb{R}$  is uniformly continuous on  $A$ , then  $f$  is continuous on  $A$ .*

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### Theorem

*If  $f : A \rightarrow \mathbb{R}$  is a Lipschitz function, then  $f$  is uniformly continuous on  $A$ .*

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### Theorem

*Let  $f : A \rightarrow \mathbb{R}$  be in  $\mathcal{R}[a, b]$ . The indefinite integral  $F(z)$  of  $f$  with basepoint  $a$  is a Lipschitz function on  $[a, b]$ .*

### Theorem (Fundamental Theorem of Calculus (Second Form))

*Let  $f \in \mathcal{R}[a, b]$  and let  $f$  be continuous at a point  $c \in [a, b]$ . Then the indefinite integral*

$$F(z) = \int_a^z f \quad z \in [a, b]$$

*is differentiable at  $c$  and  $F'(c) = f(c)$ .*

## Theorem

*If  $f$  is continuous on  $[a, b]$ , then the indefinite integral*

$$F(z) = \int_a^z f \quad z \in [a, b]$$

*is differentiable on  $[a, b]$  and  $F'(x) = f(x)$  for all  $x \in [a, b]$ .*

## Theorem (Substitution Theorem)

*Let  $J := [\alpha, \beta]$  and let  $\varphi : J \rightarrow \mathbb{R}$  have a continuous derivative on  $J$ . If  $f : I \rightarrow \mathbb{R}$  is continuous on an interval  $I$  containing  $\varphi(J)$ , then*

$$\int_{\alpha}^{\beta} (f \circ \varphi) \varphi' = \int_{\varphi(\alpha)}^{\varphi(\beta)} f.$$

## Theorem (Composition Theorem)

*Let  $f \in \mathcal{R}[a, b]$  with  $f([a, b]) \subseteq [c, d]$  and let  $\varphi : [c, d] \rightarrow \mathbb{R}$  be continuous. Then the composition  $\varphi \circ f$  belongs to  $\mathcal{R}[a, b]$ .*

## Theorem

*Suppose that  $f \in \mathcal{R}[a, b]$ . Then its absolute value  $|f|$  is in  $\mathcal{R}[a, b]$ , and*

$$\left| \int_a^b f \right| \leq \int_a^b |f| \leq M(b - a),$$

*where  $|f(x)| \leq M$  for all  $x \in [a, b]$ .*

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### Theorem (The Product Theorem)

*If  $f$  and  $g$  belong to  $\mathcal{R}[a, b]$ , then the product  $fg$  belongs to  $\mathcal{R}[a, b]$ .*

## Theorem (Integration by Parts)

Let  $F, G$  be differentiable on  $[a, b]$  and let  $f := F'$  and  $g := G'$  belong to  $\mathcal{R}[a, b]$ . Then

$$\int_a^b fG = FG \Big|_a^b - \int_a^b Fg.$$

### Theorem (Taylor's Theorem with the Remainder)

*Suppose that  $f', \dots, f^{(n)}, f^{(n+1)}$  exist on  $[a, b]$  and that  $f^{(n+1)} \in \mathcal{R}[a, b]$ . Then we have*

$$f(b) = f(a) + \frac{f'(a)}{1!}(b-a) + \dots + \frac{f^{(n)}(a)}{n!}(b-a)^n + R_n,$$

*where the remainder is given by*

$$R_n = \frac{1}{n!} \int_a^b dt f^{(n+1)}(t) \cdot (b-t)^n.$$